



An investigation on Graphical Abstracts use in scholarly articles



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ABSTRACT

Recently, Graphical Abstracts (GA) are increasingly being used in scholarly articles in order to enhance browsing and aid in paper selection. This study aims to demonstrate how GAs have been adopted in the social sciences. In the social sciences, GAs appeared for the first time in 2010, and from 2011 to 2015 a 350% increase was observed. Forty-seven percent of journals related to the social sciences have published at least one article with a GA. Among the social science disciplines, social and economic geography has most actively adopted GAs, whereas, in law, GAs are still not used. Authors use GAs to present: 1) an overview of the article, including the research process and key results (sometimes with background), 2) the key results of the study, 3) the research process or methods used, and 4) the background of the study. Authors re-use the visualizations in their manuscripts, integrate or modify the visualizations in their manuscripts, or create a new visualization for the GA. Depending on the content of GAs, different types of visualizations are used; for example, charts are mainly used to represent results and diagrams are used to present research methods or provide an overview of the article. Areas of future research into GAs are suggested.

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1. Introduction

As digital technologies have made visual information easier to generate and disseminate, visual information is becoming a more prevalent means of communication. Visualization is one of the many types of visual information, and visualization technologies have recently received attention from public and professional communities. Several definitions of visualization have been discussed, and the following are commonly included in those definitions: 1) visualization is a graphical representation of data, information, and/or knowledge, 2) the aims of visualization are to effectively convey and communicate information and/or knowledge and to amplify cognition, and 3) visualization shows what could not be presented otherwise. In addition to the values of visualization itself, it has been found that when verbal and visual techniques are combined to convey information, attention, understanding, and recall of the information are enhanced. The effectiveness of combining text and visual information has been investigated experimentally and shown in fields such as education, public health, and marketing.

Journal articles have long integrated visual information, such as tables, charts, graphics, and photos, to effectively present

results and to show models/frameworks, examples, and contextual backgrounds within scholarly communications. Recently, Elsevier initiated the “Article of the Future” project,¹ and the Graphical Abstract (GA), which utilizes visual information as an abstract, is a part of the project. The Article of the Future project “aim[s] to revolutionize the traditional format of the academic paper.” The project has developed discipline-specific prototypes which enhance navigation and browsing, add content (e.g., interactive tables, figures, and maps, graphical abstracts, etc.), and provide context (e.g., related information, related articles, etc.).

Among these features of the Article of the Future project, the focus of the current study is the Graphical Abstract (GA). The Elsevier website provides a working definition of GA: “[A] Graphical Abstract is a single, concise, pictorial and visual summary of the main findings of the article. [...] This could either be the concluding figure from the article or a figure that is specially designed for the purpose, which captures the content of the article for readers at a single glance.”² GAs in Elsevier journals do not appear in PDF file or print format, only in online articles and online search result lists. Therefore, as addressed on the Elsevier website, a GA is intended to

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¹ <http://www.articleofthefuture.com/>.

² <http://www.elsevier.com/authors/journal-authors/graphical-Abstract>.

enhance browsing and aid in paper selection by allowing a reader to quickly understand the main focus of the paper.³

Although the benefits of GAs can be anticipated based on the general benefits of visualization and multimodal communication, there has been a lack of research exploring how GAs are practically utilized in scholarly articles and what their influences are on scholarly communication. In this context, this study attempts to initiate research on GAs and establish a future research agenda. GAs are popularly used in science and engineering journals as a way of representing, for example, molecular geometry, cell structures, mechanical structures, and so on. Although GAs are less popular in the social sciences than in the hard sciences and engineering disciplines, the implications from research in the social sciences have a greater potential to be applied to other knowledge creation and knowledge sharing cases involving humans and society in general. Therefore, this study aimed to understand how GAs are employed in the social sciences. The following research questions are addressed:

RQ1: How frequently are Graphical Abstracts adopted in social science disciplines?

RQ2: What types of Graphical Abstracts are used in social science disciplines?

This study will demonstrate how GAs are adopted in the social sciences and suggest how to enhance effectiveness of Graphical Abstract for scholarly communication.

2. Related studies

2.1. Visualization

As Bresciani stated, “visual representations are used to organize and structure information and ideas in order to convey knowledge to amplify cognition and to enhance communication” (Bertschi et al., 2011, p. 14). It has been demonstrated that visualization is valuable for browsing, exploring, developing insights, and making new discoveries (Fekete, Wijk, Stasko, & North, 2008). Visualization is an interdisciplinary field, and terminologies of data-, information-, and knowledge visualizations have been variously employed, according to the contexts and applications of the visualization. Mayer (2010) attempted to compare these three types of visualizations using the concepts of data, information, and knowledge. Burkhard (2005) compared information visualization and knowledge visualization in terms of ten aspects, goals, benefits, content, recipients, influence, proponents, contribution, root, means, and complementary visualizations.

Some studies have presented taxonomies of visualizations. A periodic table of visualization methods (Lengler & Eppler, 2007) systematically categorized 100 visualization methods into six groups: data visualization (visual representation of quantitative data in schematic form), information visualization (interactive visual representations of data), concept visualization (representation of qualitative concepts), metaphor visualization (utilization of visual metaphors), strategy visualization (systematic use of visual representations for strategies), and compound visualization (combinations of different types of visualization). In addition, each visualization method was thought of in three dimensions: task and interaction (overview, detail and overview, and detail), cognitive processes (convergent thinking and divergent thinking), and represented information (structure and process). Eppler and

Burkhard (2005) organized types of knowledge visualizations as structured text and tables (organization of knowledge with grids), heuristic sketches (drawings facilitating knowledge transfer and creation of insight), conceptual diagrams (schematic representation of abstract ideas in standardized formats), visual metaphors (utilization of visual metaphors to enhance knowledge transfer), knowledge maps (graphical representations using cartographic conventions), and interactive visualizations and animations (visualizations which allow users to control and interact with the visualizations).

2.2. Multimodal information

Whereas studies on visualization focus on visual information itself, other research has established theories as to how humans process text-visual information. These theories are based on the assumption that “people learn more deeply from words and pictures than from words alone” (Mayer, 2005, p. 31). Dual coding theory, posited by Paivio (1986), assumes that there are two different cognitive subsystems, one for verbal information and the other for non-verbal information, and that these two subsystems separately process and create representations of the corresponding type of information. Dual coding theory posits that the two subsystems operate independently from one another but frequently work simultaneously, and, therefore, a multimodal stimulus can foster greater recall and understanding than can a unimodal stimulus. Mayer's (2005) cognitive theory of multimedia learning (CTML), which is based on dual coding theory, includes the notion that verbal and visual presentations, which enter the mind through sensory memory, are processed through selection, organization, and integration processes. During the selection process, relevant words and/or images are selected from the presented texts and/or illustrations, and in the organization process, selected words and images form coherent verbal and/or visual representations. The selection and organization processes are conducted in two separate channels. Then, the verbal and visual representations are integrated with prior knowledge retrieved from long-term memory. Selection, organization, and integration processes occur in working memory.

Although these theories support the superiority of multimodal information, it is not the case that multimodality is always beneficial or equally as effective as other means of presentation. There are lines of research which have developed a set of principles for effective visual representation, particularly in learning environments. Mayer (2005) found that human cognitive capacity is easily overloaded by an excess of both essential and extraneous materials, and presented several principles which can reduce extraneous overload; the principles include coherence, signaling, and spatial contiguity. The coherence principle suggests exclusion of extraneous materials. Sung and Mayer (2012)'s study supported the coherence principle by demonstrating that graphics which are directly related to an instructional goal enhance cognitive processing, whereas irrelevant graphics do not enhance learning, although all graphics, even if they are irrelevant, have a positive effect on affective measures. The spatial contiguity principle recommends placing text and visuals that correspond with one another close together. Johnson and Mayer (2010) supported this principle by comparing eye movements of subjects viewing a slide with four different versions of text/image placement. The signaling principle recommends providing cues that will direct attention, and includes how to organize/process presented information. Holsanova, Holmberg, and Holmqvist (2009) found that people process complex materials more easily when guidance is provided through a spatially and conceptually organized layout.

Eitel, Scheiter, Schüler, and Nyström (2013) examined the scaffolding effect, which occurs even following a brief exposure to spatial information extracted from an image; the scaffolding effect

³ “A Graphical Abstract should allow readers to quickly gain an understanding of the main take-home message of the paper and is intended to encourage browsing, promote interdisciplinary scholarship, and help readers identify more quickly which papers are most relevant to their research interests. Authors must provide an image that clearly represents the work described in the paper.” (<http://www.elsevier.com/authors/journal-authors/graphical-abstract>.)

Table 1
Coding scheme for the use of visualization.

Type of visualization
• Table – Arrangements of information with rows and columns • Chart – Visual representation of numeric data using standard quantitative formats, such as bar, scatter plot, and pie charts • Geo Map – Geographical map, whether it is photo or drawing • Data Map – Data visualization, mostly for large amounts of data, such as network, heat map, etc. • Diagram – Simplified and structured representation of qualitative information or abstract ideas using standardized shapes, such as lines, boxes, circles, arrows. • Graphic – Pictorial representation including drawing, cartoon, and illustration. • Photo – An image of an object, scene, person, etc., mostly made by a camera • Combination – A visualization is composed of different types of visualizations
Content of visualization
• Background: Visualization of contexts, background, and introductory information • Method: Visualization of research design and process • Results: Visualization of study results, discussion and conclusion • Overview: Visualization covering overall research, such as demonstration of research process and results • Others: Visualization representing other aspects of manuscript
Relation to the manuscript
• New: GA is newly created • Duplicated: GA is copied from an existing visualization in the manuscript • Modified: GA is modified from an existing visualization in the manuscript • Integrated: GA is created by integrating existing visualizations in the manuscript

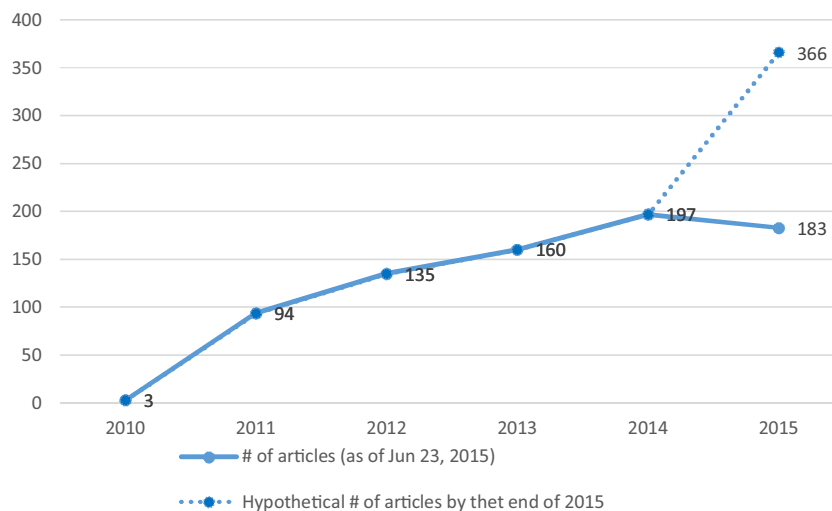


Fig. 1. The number of articles with a GA by year.

is “that spatial information [is] reactivated and used as a mental scaffold” when people read subsequent text (p. 48). In addition, studies have investigated the effect of the complexity of visual information, such as simplified vs. complex and concrete vs. abstract, and the results showed that the influence of different types of images varies depending on the context in which and the purposes for the images being used. In [Butcher \(2006\)](#)’s study, simplified diagrams generated an improvement in understanding and memorization of text than did complex diagrams. Concrete images, which are closely related to the real world, are more easily incorporated into long-term memory and enhance learning ([Koedinger & Nathan, 2004](#)), whereas abstract images are more useful for understanding structural features of the content at hand ([Colin, Chauvet, & Viennot, 2002](#)).

As discussed, there are lines of research on the use of visualizations and visual information for enhancing communication, but there have been only a few studies addressing the ways in which visual information is used for scholarly communications. [Friedman \(2014\)](#) examined the relationship between research methods and type of visualization used to present research findings by analyzing a conference proceeding published between 1990 and 2010.

[Healy and Moody \(2014\)](#) analyzed how visualizations are used in sociology to present the results of statistical modeling.

3. Methodology

On June 23, 2015, using the advanced search menu from Science Direct, Elsevier journal articles, which are categorized in the Social Sciences and embed a GA, were retrieved. A total of 1261 journal articles were retrieved from 110 journals. As of 2015, there were 234 Elsevier journals categorized as social science journals.⁴; therefore, 47% of social science journals included at least one article with an embedded GA.

Articles in Science Direct have keywords, but there is no systematic way of mapping those keywords into disciplines. So, to analyze the use of GAs by discipline; we utilized the subject category of Web of Science.

In order to analyze the use of GAs in terms of discipline, Web of Science Subject Categories were retrieved from each of 110 jour-

⁴ <http://info.sciencedirect.com/techsupport/journals/sc2015.htm>.

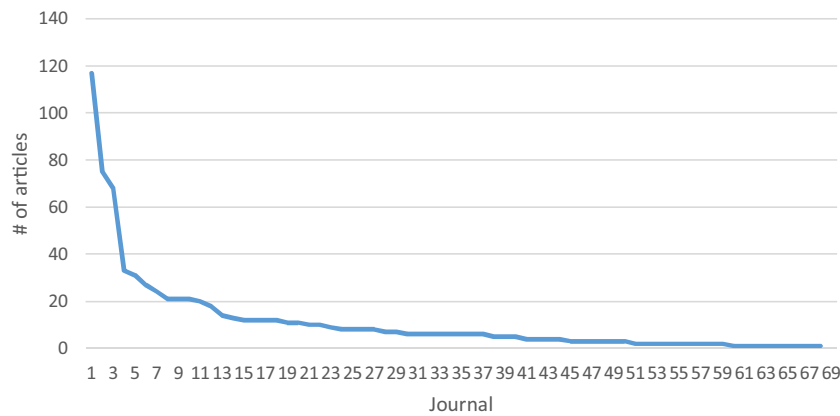


Fig. 2. The number of articles with a GA per journal.

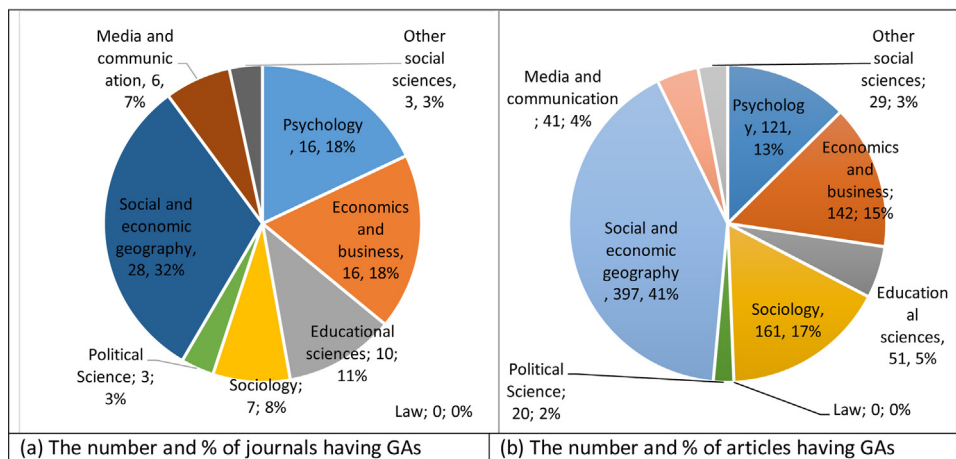


Fig. 3. Numbers and percentages of journals and articles with GAs by discipline.

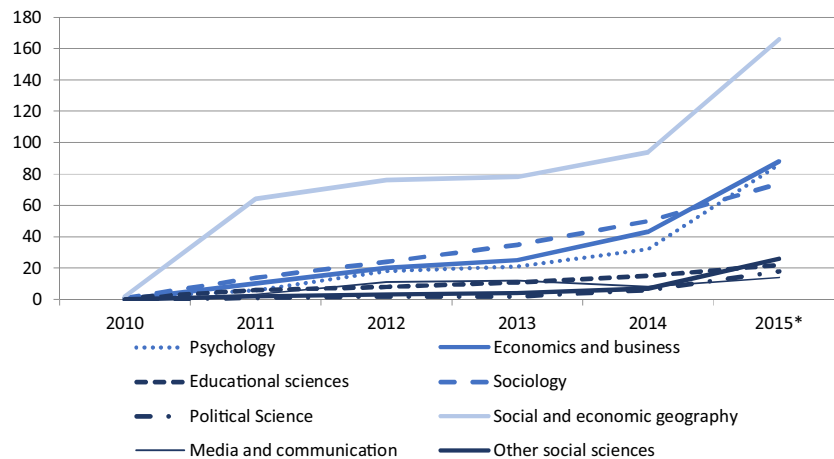


Fig. 4. The number of articles with GAs by year and discipline.

*The number of articles in 2015 was hypothetically generated by doubling the number in the current dataset, because the data was collected up until June of 2015.

nals. The Web of Science subject categories include 248 subject terms which cover all disciplines, including the humanities, science and engineering, and social sciences⁵ Then, the mapping table

between Web of Science subject category and the OECD Category Scheme was employed.⁶ The OECD Category Scheme is composed of six major classes (natural sciences, engineering and technology, medical and health sciences, agricultural sciences, social sciences,

⁵ Web of Science Category Terms <https://images.webofknowledge.com/WOKRS57B4/help/WOS/hp.subject.category.terms.tasca.html> (retrieved on July 7, 2015).

⁶ <http://ipsience-help.thomsonreuters.com/incitesLive/globalComparisonsGroup/globalComparisons/subjAreaSchemesGroup/oecd.html>.

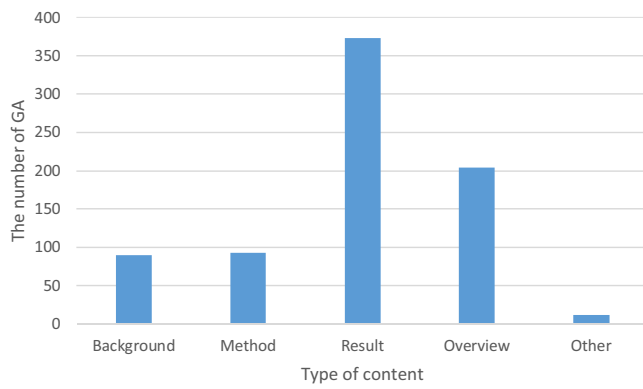


Fig. 5. Content of GAs.

and humanities), along with 42 minor classes. The social science class has 9 subclasses (psychology, economics and business, educational sciences, sociology, law, political science, social and economic geography, media and communications, and other social sciences). During this process, 42 journals were excluded from the dataset for two reasons: 1) 18 journals were not included in the Web of Science, and 2) 24 journals were not classified as social sciences according to the Web of Science Subject Categories and OECD Category Scheme. As a result, a total of 772 articles from 68 journals were categorized into the 9 subclasses of social science.

The use of visualizations in GAs was analyzed in terms of type, content, and relationship to the manuscripts. A preliminary coding scheme was developed with 10% of data sets (78 articles). Lengler and Eppler's (2007) and Eppler and Burkhard's (2007) periodic table of visualization and examples provided a basis for the development of the preliminary coding scheme. After the preliminary coding scheme was established, another 10% of data sets were coded using that preliminary coding scheme, and minor revisions were made (see Table 1). The entire data set was coded by one author, and another author coded 10% of the data set in order to assess inter-coder reliability. The average ratio of inter-coder reliability is 93% (91% for type of visualization, 96% for content of visualization, and 92% for relationship to the manuscript).

4. Findings

4.1. The number of Graphical Abstracts

The distribution of the 772 articles embedding GA by publication year are shown in Fig. 1. Since the data set was collected in June 2015, the hypothetical number of articles for 2015 was generated by doubling the number of articles published as of June 2015. With the hypothetical number, from 2011 to 2015, the number of GAs has increased approximately 350% in the social sciences as a whole.

Fig. 2 shows the number of articles with a GA per journal. One journal had more than 100 articles with a GA (*Journal of Archaeological Sciences*: 117). Two journals had more than 50 articles with a GA (*Land Use Policy*: 75, *Landscape and Urban Planning*, 68). 20 journals had one or two articles with a GA.

Fig. 3 presents the distributions of 68 journals and 772 articles in terms of discipline. Distributions of journals and articles show similar patterns, although there are a few differences. Overall, social and economic geography most actively use GAs, followed by economics and business, psychology, educational sciences, sociology, media and communication, and political science. The area of law did not have any articles embedding GAs.

Fig. 4 demonstrates the increasing trend of GA usage in each discipline. Social and economic geography showed a remarkable increase in GA usage. Economics and business, psychology, and sociology belong to the second group, and educational sciences, political science, and media and communication are the disciplines which least often use GAs.

4.2. The use of Graphical Abstracts

The use of GAs was analyzed by content. Among 772 articles, 48.32% represented the results of the article in the GA, 26.42% presented an overview of the article, 12.05% presented research methods, and 11.66% provided background or contextual information (Fig. 5).

Regarding types of visualization, in general, diagram, combination, and photo were often used as GAs. Depending on the content of GAs, different types of visualizations were preferred (Fig. 6). Whereas a diagram was used to visualize all types of article con-

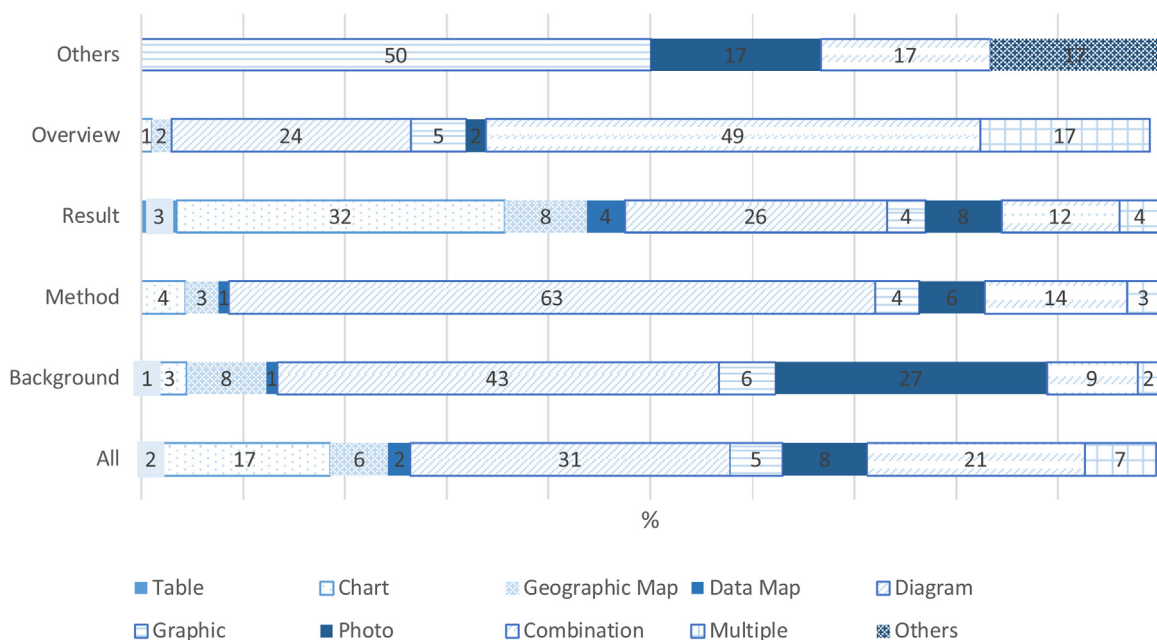


Fig. 6. Type of visualization used in GAs.

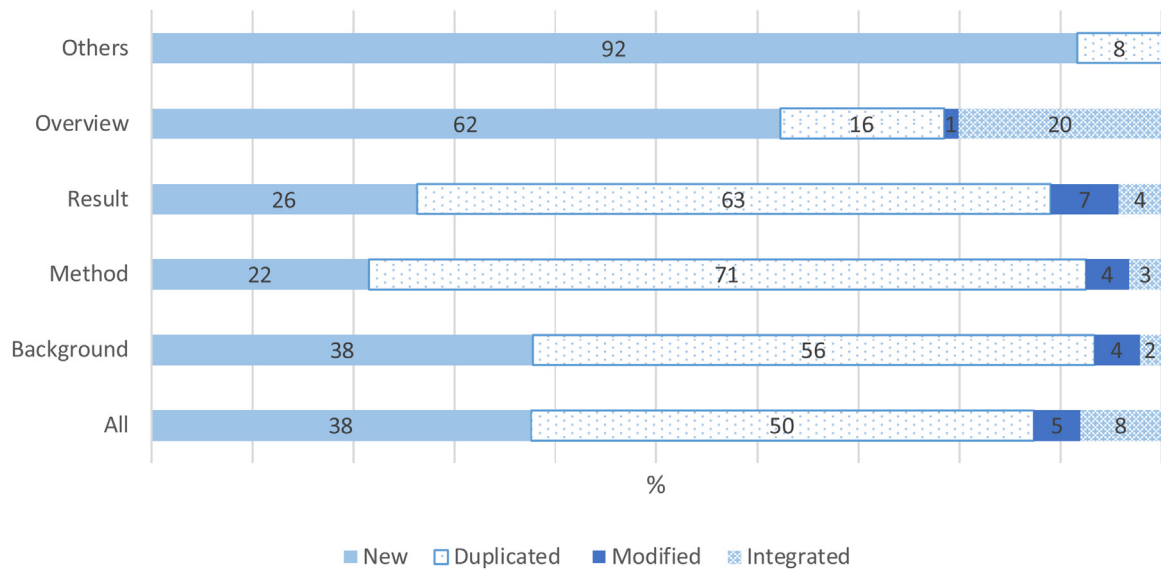


Fig. 7. Methods of generating GAs.

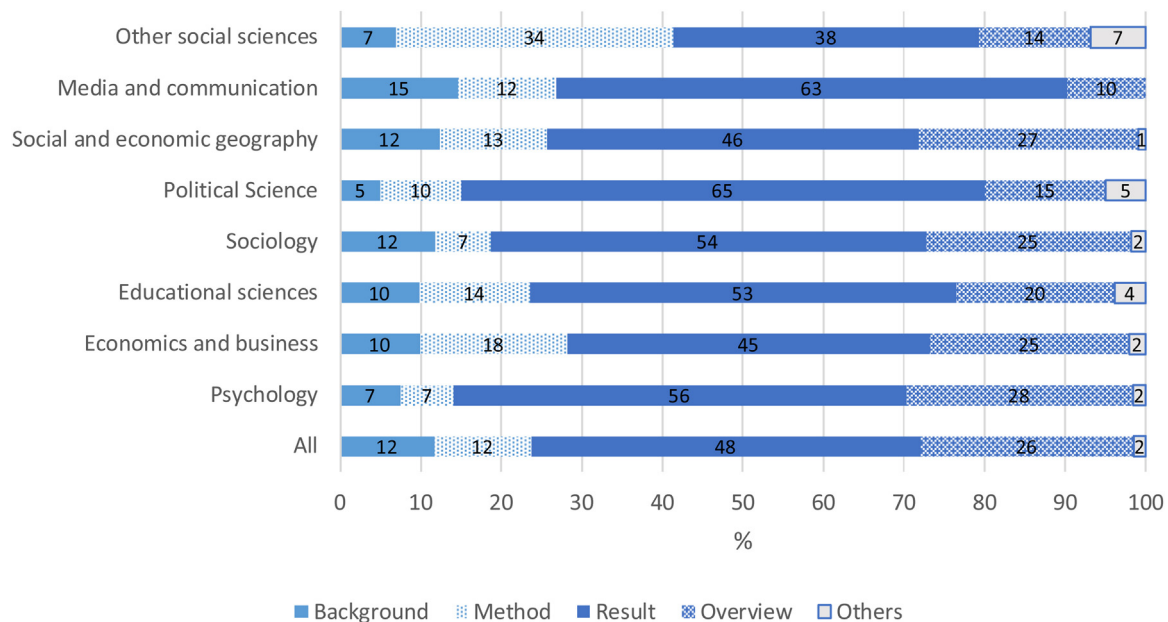


Fig. 8. Content of GAs in terms of discipline.

tent, charts were mostly used to present results, combinations were used to provide an overview of the article, and photos were used to depict background and results. As presented in Fig. 7, 38% of GAs were newly created, whereas 50% were simply duplicated from a manuscript's existing visual components. Five percent were modified from existing visual components, and 8% were generated by integrating existing visual components. Among the GAs giving an article overview, more than 60% were newly created and 20% were generated by integrating existing visualizations in the manuscripts, whereas of the GAs representing results and methods, 63% and 71%, respectively, were simply duplicated.

Depending on type of content, certain characteristics of visualizations were identified. First, the most frequently used types of visualizations in GAs presenting background information were diagrams (43%) and photos (27%). 38% of GAs representing background information were new visualizations and 56% of them were

duplicated visualizations. Diagrams were often used to show conceptual frameworks from previous studies, and, in these cases, most of them were duplicated visualizations. Photos of relevant objects or scenes were often used to provide readers with background information. Second, a dominant type of visualization for GAs representing methods was the diagram, which shows methodological processes. 71% of GAs presenting research methods were duplicated and 22% of them were new. Third, charts (32%) and diagrams (26%) were often used to represent results. Charts were used to quantitatively represent key results, by duplicating an existing chart. Diagrams were used to present frameworks of conclusions drawn from the study results. Although 63% of GAs depicting results were duplicated, sometimes, new charts and diagrams were created. In these cases, the results were most often presented as tables in the manuscript, but charts and diagrams were newly created to grab more of the readers' attention. Fourth, most GAs representing

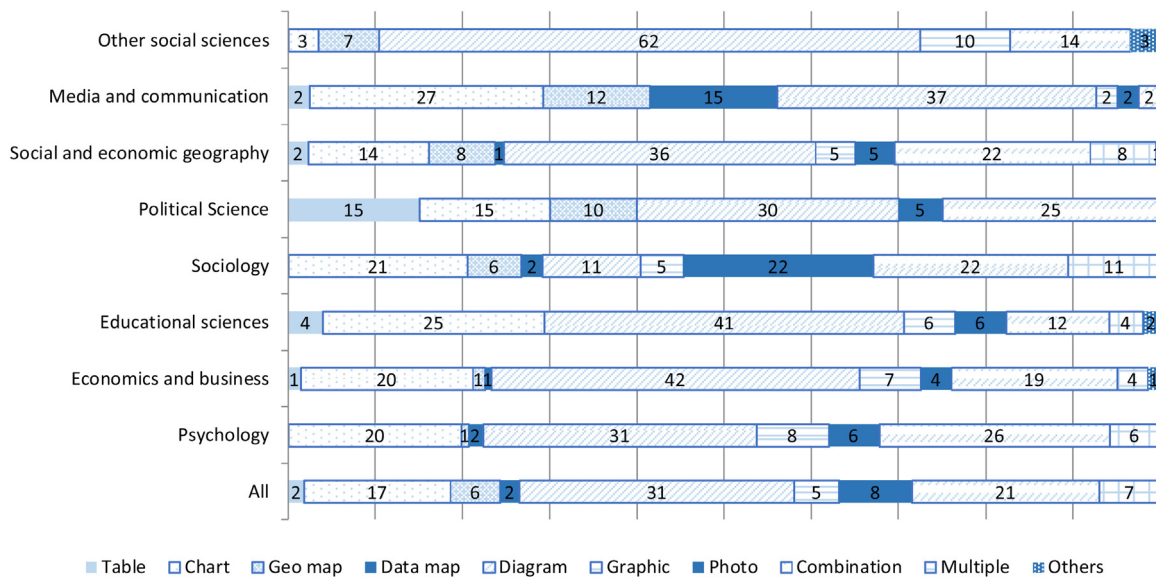


Fig. 9. Type of visualizations in terms of discipline.

overviews were newly created (62%), or were created by integrating existing visualizations (20%). These GAs presented methods and results, but sometimes they also demonstrated background, methods and results. Combinations (49%), multiple visualizations (17%), and diagrams (24%) were often used to provide overviews. Finally, there were a few GAs which were categorized as “others,” with a common example being graphics (50%). Some GAs in this category presented irrelevant visualizations, but others attempted to present the overall themes or messages of the studies. For example, an article which studied the role of social media in violence amongst youths included a graphic of a gun as a GA.

Fig. 8 shows the content of GAs in terms of discipline. In almost every discipline, results were frequently used as GA content, followed by overview. In media and communication and political science papers, the percentage of GAs that depict results was higher than in other disciplines, and the percentage presenting an overview was lower. In economics and business, the percentage presenting methods was relatively higher than in other areas. Regarding types of visualization (Fig. 9), as addressed above, overall, a diagram was the dominant type, followed by combination, chart, and photo. Sociology showed a unique pattern, in which photos are most frequent, followed by charts and diagrams. Some types of visualization were more often used in specific fields. Diagrams were more often used in economics and business and educational sciences, and charts were more often used in media and communication and educational sciences. The use of photos in sociology, the use of tables in political science, and the use of data maps in media and communication were noticeable. The use of a Geo map was greater in the areas of media and communication, political science, and social and economic geography.

5. Discussion

Scholarly articles attempt to effectively communicate their findings through optimized, standardized formatting, which generally includes title, abstract, instruction, literature review, methods, results, discussion, conclusion, and reference sections (Somersault18:24, [n.d.]).⁷ With the development of digital technologies,

better formats for scholarly communication are being explored, and utilization of GAs is one such attempt at improvement. There has been a substantial history of research on graphics in text and the effectiveness of visual communication, but it is rare in the field of scholarly communication. This study attempts to initiate research on GAs.

In the social sciences, GA was adopted for the first time in 2010. Although the use of GAs is not as popular as it is in science fields, the recent increase in GA usage is remarkable; from 2011 to 2015, a 350% increase was observed. This trend justifies the needs for more attention to GAs in scholarly communications. Multimodality studies can be viewed from two aspects, production and reception (Holsanova, 2012), and GAs can be examined from these two perspectives.

First, from the production perspective, this study examined the types of content and types of visualizations that GAs adopt. We demonstrated that the authors of scholarly articles have attempted to use GAs to present 1) an overview of the articles, including both the research process and key results (sometimes with the addition of background information), 2) major findings of the study, 3) research process or methods, and 4) background of the study. The content of GAs can be considered from three aspects, for their improvements. First, a GA, by itself, can help readers to immediately understand key points of an article, and, furthermore, those key points can guide readers while they read the articles and facilitate their understanding of presented information. Eitel et al. (2013) demonstrated a scaffolding effect, in which even a brief exposure to spatial information extracted from an image is useful as a mental scaffold when people read subsequent text. GAs can play the same role for the readers of articles, and the precise type of content that can effectively facilitate users' understanding while reading articles should be examined. Second, a GA can help readers by interacting with other journal components. Journal articles use traditional mechanisms to convey key points, such as abstracts, keywords, and, recently, highlights. As Conner-Simons (2015) asserted that people spend more time on the text, even when there are great visualizations, it is assumed that although a GA is valuable by itself, readers would still spend more time on the abstracts, keywords, and highlights in order to extract key information. Using dual coding theory (Paivio, 1986) and CTML (Mayer, 2005), it has been shown that a combination of text and visuals enhances readers' cognitive processing, but are not equally effective in all cases. Therefore, GA use

⁷ <http://www.somersault1824.com/a-graphical-abstract-opens-the-door-to-your-research-publication/>.

should be considered based on an understanding of how GAs interact with traditional mechanisms. Third, this study demonstrated different features of GA usage depending on discipline. Social economics and geography are the disciplines which most actively use GAs, whereas there was no GA use in law-related articles; political science and media and communication used GAs to present results more often than did other disciplines; and educational science and economics and business used diagrams more often than did other disciplines. It is necessary, then, to examine the possible relationships between disciplines and use of GAs.

The current study found that, depending on the content of GAs, different types of visualizations were used; for example, charts were mainly used to represent results, and diagrams were used to present research methods or give an overview of the articles. Some studies have elaborated on the best ways to present visual information for more effective communication, such as how to create charts and tables (Few, 2004), how to create visual information (Tufte, 1991, 1997), which type of diagram (concrete vs. abstract) is more effective in facilitating understanding (Hegarty, Carpenter, & Just, 1991; Moreno, Ozogul, & Reisslein, 2011; Perry, Howard, & Tracey, 1990), how to place text and visualizations for more effective cognitive processing (Holsanova et al., 2009; Salway & Martinec, 2005), and so on. Although these studies demonstrated characteristics of effective visualizations, there is no evidence that the authors of scholarly articles are acquainted with those visual design principles. As Mason, Pluchino, Tornatora, and Ariasi (2013) stated, “Simply adding pictures to texts does not automatically lead to better learning.” (p. 357). Therefore, it would be desirable to provide authors with the characteristics of effective visualizations.

Second, design of GAs should be based on a thorough understanding of readers' cognitive processes. By definition, GAs are intended to enhance recipients' ability to quickly browse and select relevant journal articles. Currently, readers can only view GAs online. However, there is evidence from previous studies on visual communication that GAs may have greater potential than just browsing and selection. Studies on visual or multimodal materials used in education (Carney & Levin, 2002; Hannus & Hyönä, 1999) have shown that GAs can enhance understanding, retention, and recall of the information presented in articles. Therefore, it is necessary to examine how GAs operate during the processes of browsing, selection, and understanding of scholarly articles, and how GAs enhance the effectiveness of those tasks. In addition, it is necessary to investigate how readers integrate information from GAs with that from other parts of the manuscripts. The means of integrating information from GAs and other parts of manuscripts may differ depending on readers' levels of domain-related knowledge, language skills, and purpose for reading the manuscript. More thorough investigation into readers' cognitive behaviors would provide guidance in producing effective GAs.

This is an initial study on GAs; the uses of GAs were analyzed by conducting content analysis of GAs. Although the patterns of GA use were revealed by the current research, one limitation of the current study is that qualitative measures are missing. In order to explain those patterns, research into the perceptions and behaviors of scholars and researchers, as producers as well as receivers of GAs, should be conducted. Another future research area is to examine influences of GAs in scholarly communication. Considering the purpose of a GA, which is to enhance browsing and assist in selection of relevant articles, the authors compared the numbers of citations between articles with GAs and articles without GAs. This simple comparison found no evidence that GAs influence citations. Given the current study, it is difficult to explain this result, but further research investigating recipients' cognitive processes would be able to address the issue. Also, further research which measures influences of GAs in a more systematic way may find other results.

6. Conclusion

In the disciplines of the social sciences, the use of GAs is not popular at this point in time; however, the recent increase in GA usage is remarkable. GAs appeared for the first time in 2010, and from 2011 to 2015, a 350% increase in GA use was observed. Among the social science disciplines, social and economic geography is the subspecialty which has most actively adopted GAs. Economics and business, psychology, and sociology showed higher usage of GAs than did educational sciences, political science, and media and communication. In law, no use of GAs was found. The authors of scholarly articles attempted to use GAs to present 1) an overview of the articles, including the research process and key results (sometimes with background information), 2) key results of the study, 3) research processes or methods, and 4) the background of the study. Authors re-use the visualizations in their manuscripts, integrate or modify the visualizations in their manuscripts, or create a new visualization for use as a GA. Depending on the content of GAs, different types of visualizations were used; for example, charts were mainly used to represent results and diagrams were used to present research methods or an overview of the articles.

This study is an initial foray into research on the use of GAs in social sciences, and it suggests the following questions for future research:

- How do readers cognitively process GAs?
- How do readers integrate GAs with other parts of the manuscript?
- How do GAs enhance the effectiveness of scholarly communications?
- What are the characteristics of effective GAs?
- Are there factors (demographics, tasks, etc.) which influence the effectiveness of GAs?
- What are the motivations of GA creators?
- What are the influences of GAs in article citation and use?

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